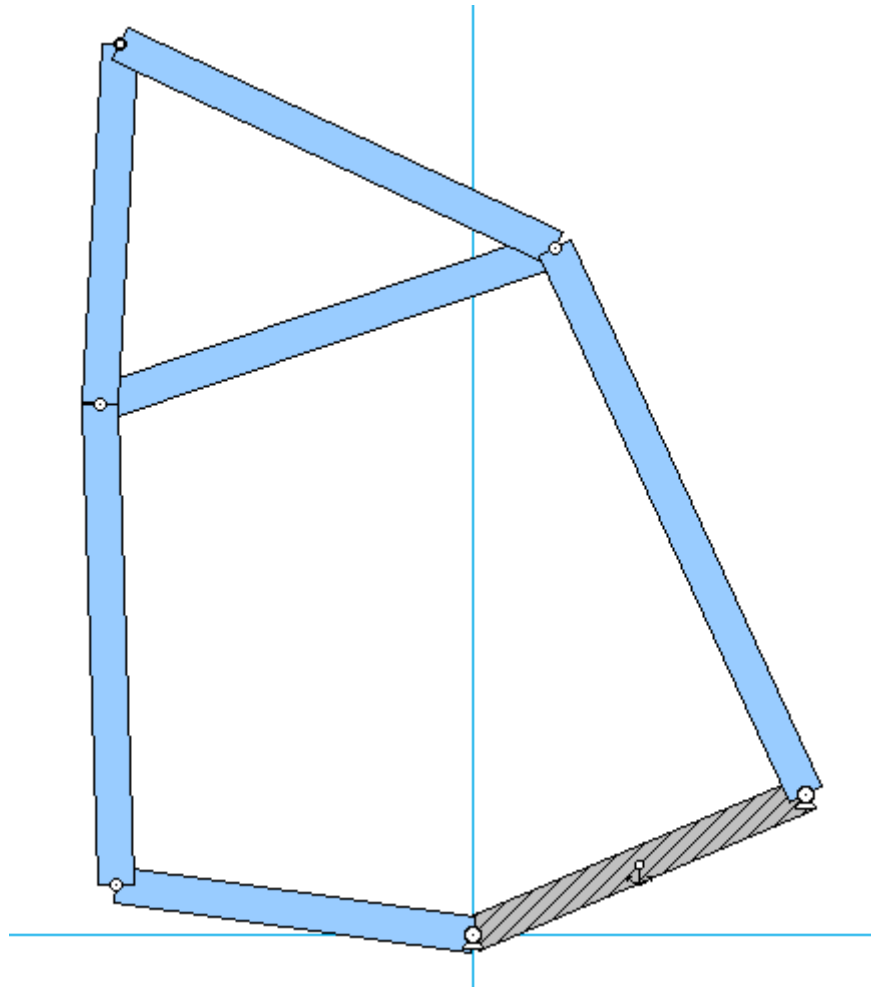
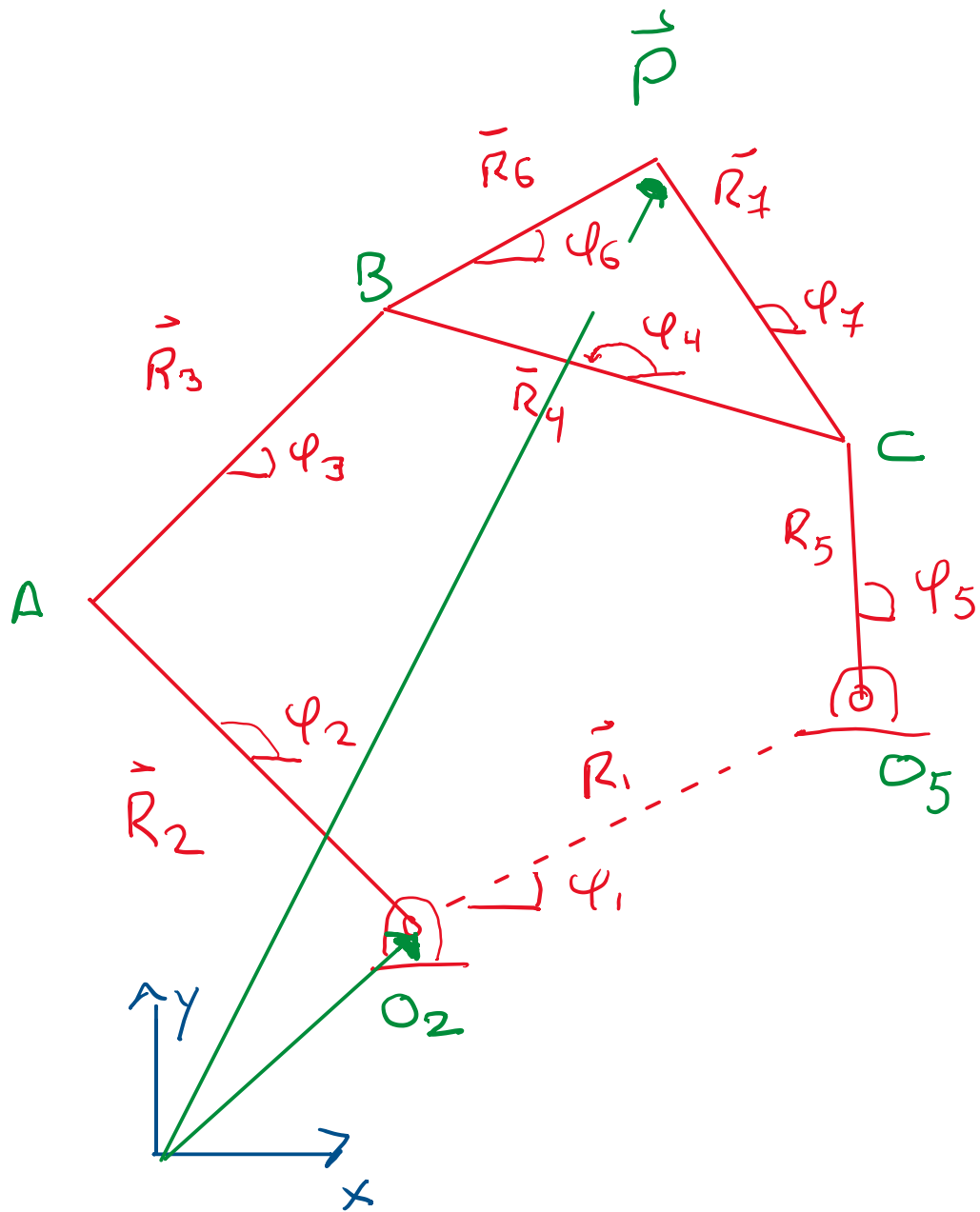


# Mecanismo 5 barras – 2 GdL



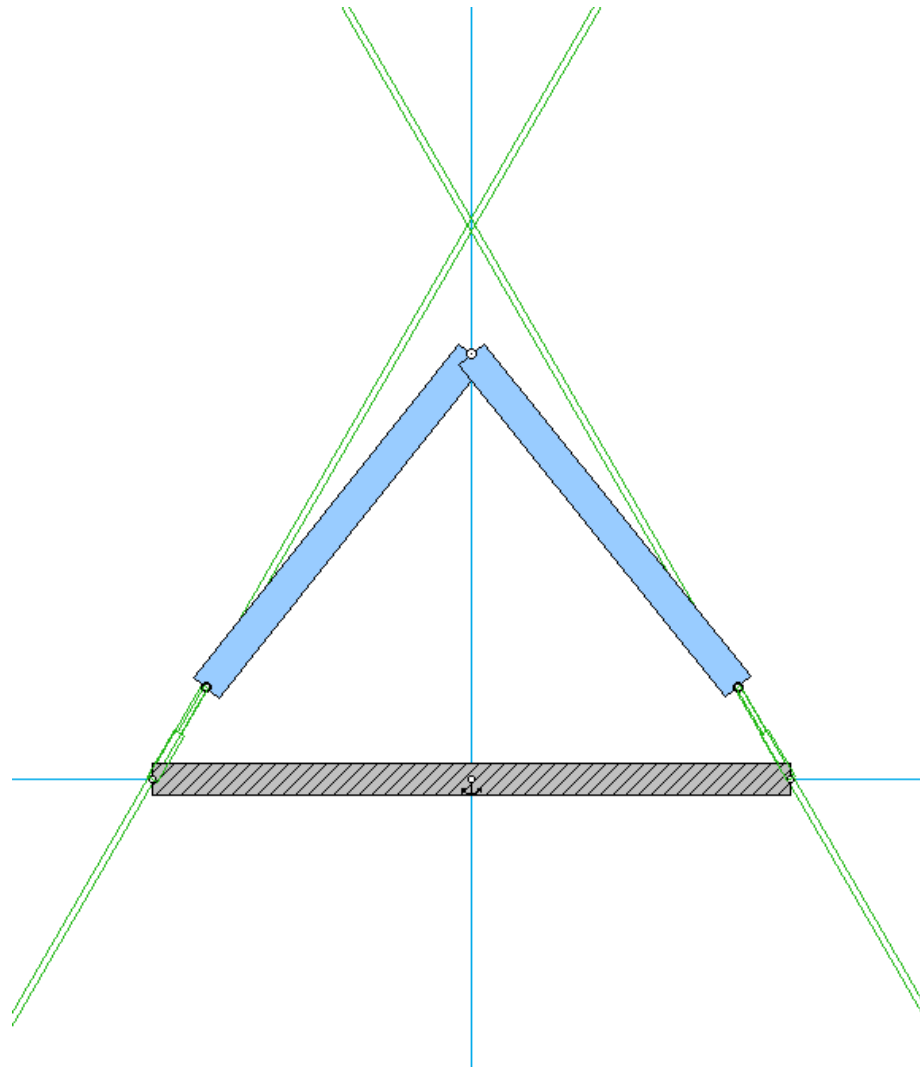


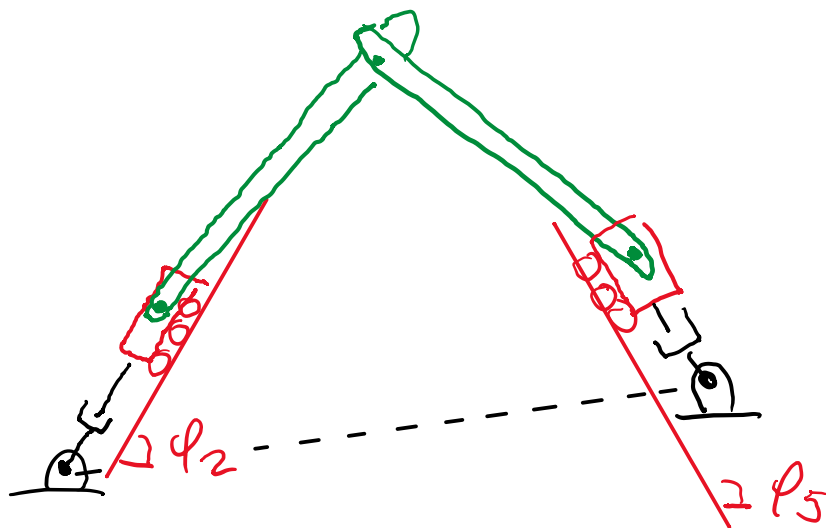
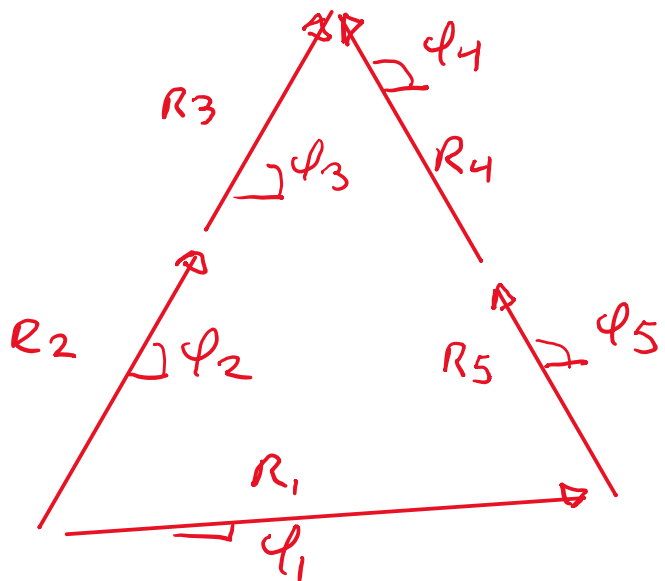
$$\Phi(\mathbf{q}, t) = \begin{bmatrix} -\mathbf{r}_1 + \mathbf{r}_2 + \mathbf{r}_3 - \mathbf{r}_4 - \mathbf{r}_5 \\ \mathbf{r}_4 + \mathbf{r}_6 - \mathbf{r}_7 \\ \mathbf{o}_2 + \mathbf{r}_2 + \mathbf{r}_3 + \mathbf{r}_6 - \mathbf{p} \\ \varphi_2 - \varphi_{20} - \omega_{20}t - \frac{1}{2}\alpha_{20}t^2 \\ \varphi_5 - \varphi_{50} - \omega_{50}t - \frac{1}{2}\alpha_{50}t^2 \end{bmatrix} = \begin{bmatrix} -r_1c_1 + r_2c_2 + r_3c_3 - r_4c_4 - r_5c_5 \\ -r_1s_1 + r_2s_2 + r_3s_3 - r_4s_4 - r_5s_5 \\ r_4c_4 + r_6c_6 - r_7c_7 \\ r_4s_4 + r_6s_6 - r_7s_7 \\ o_{2x} + r_2c_2 + r_3c_3 + r_6c_6 - p_x \\ o_{2y} + r_2s_2 + r_3s_3 + r_6s_6 - p_y \\ \varphi_2 - \varphi_{20} - \omega_{20}t - \frac{1}{2}\alpha_{20}t^2 \\ \varphi_5 - \varphi_{50} - \omega_{50}t - \frac{1}{2}\alpha_{50}t^2 \end{bmatrix} \quad \mathbf{q} = \begin{bmatrix} \varphi_2 \\ \varphi_3 \\ \varphi_4 \\ \varphi_5 \\ \varphi_6 \\ \varphi_7 \\ p_x \\ p_y \end{bmatrix}$$

$$\mathbf{J}(\mathbf{q}, t) = \begin{bmatrix} -r_2s_2 & -r_3s_3 & r_4s_4 & r_5s_5 & 0 & 0 & 0 & 0 \\ r_2c_2 & r_3c_3 & -r_4c_4 & -r_5c_5 & 0 & 0 & 0 & 0 \\ 0 & 0 & -r_4s_4 & 0 & -r_6s_6 & r_7s_7 & 0 & 0 \\ 0 & 0 & r_4c_4 & 0 & r_6c_6 & -r_7c_7 & 0 & 0 \\ -r_2s_2 & -r_3s_3 & 0 & 0 & -r_6s_6 & 0 & -1 & 0 \\ r_2c_2 & r_3c_3 & 0 & 0 & r_6c_6 & 0 & 0 & -1 \\ 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \end{bmatrix} \quad \dot{\phi}_t = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ -\omega_{20} - \alpha_{20}t \\ -\omega_{50} - \alpha_{50}t \end{bmatrix}$$

$$\dot{\mathbf{J}}(\mathbf{q}, t) = \begin{bmatrix} -r_2\omega_2c_2 & -r_3\omega_3c_3 & r_4\omega_4c_4 & r_5\omega_5c_5 & 0 & 0 & 0 & 0 \\ -r_2\omega_2s_2 & -r_3\omega_3s_3 & r_4\omega_4s_4 & r_5\omega_5s_5 & 0 & 0 & 0 & 0 \\ 0 & 0 & -r_4\omega_4c_4 & 0 & -r_6\omega_6c_6 & r_7\omega_7c_7 & 0 & 0 \\ 0 & 0 & -r_4\omega_4s_4 & 0 & -r_6\omega_6s_6 & r_7\omega_7s_7 & 0 & 0 \\ -r_2\omega_2c_2 & -r_3\omega_3c_3 & 0 & 0 & -r_6\omega_6c_6 & 0 & 0 & 0 \\ -r_2\omega_2s_2 & -r_3\omega_3s_3 & 0 & 0 & -r_6\omega_6s_6 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix} \quad \ddot{\phi}_t = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ -\alpha_{20} \\ -\alpha_{50} \end{bmatrix}$$

# Mecanismo 5 barras corredera – 2 GdL





$$\boldsymbol{\Phi}(\mathbf{q},t)=\begin{bmatrix} -\mathbf{r}_1+\mathbf{r}_2+\mathbf{r}_3-\mathbf{r}_4-\mathbf{r}_5 \\ r_2-r_{20}-v_{20}t-\frac{1}{2}a_{20}t^2 \\ r_5-r_{50}-v_{50}t-\frac{1}{2}a_{50}t^2 \end{bmatrix}=\begin{bmatrix} -r_1c_1+r_2c_2+r_3c_3-r_4c_4-r_5c_5 \\ -r_1s_1+r_2s_2+r_3s_3-r_4s_4-r_5s_5 \\ r_2-r_{20}-v_{20}t-\frac{1}{2}a_{20}t^2 \\ r_5-r_{50}-v_{50}t-\frac{1}{2}a_{50}t^2 \end{bmatrix}$$

$$\mathbf{q}=\begin{bmatrix} r_2 \\ r_5 \\ \varphi_3 \\ \varphi_4 \end{bmatrix}$$

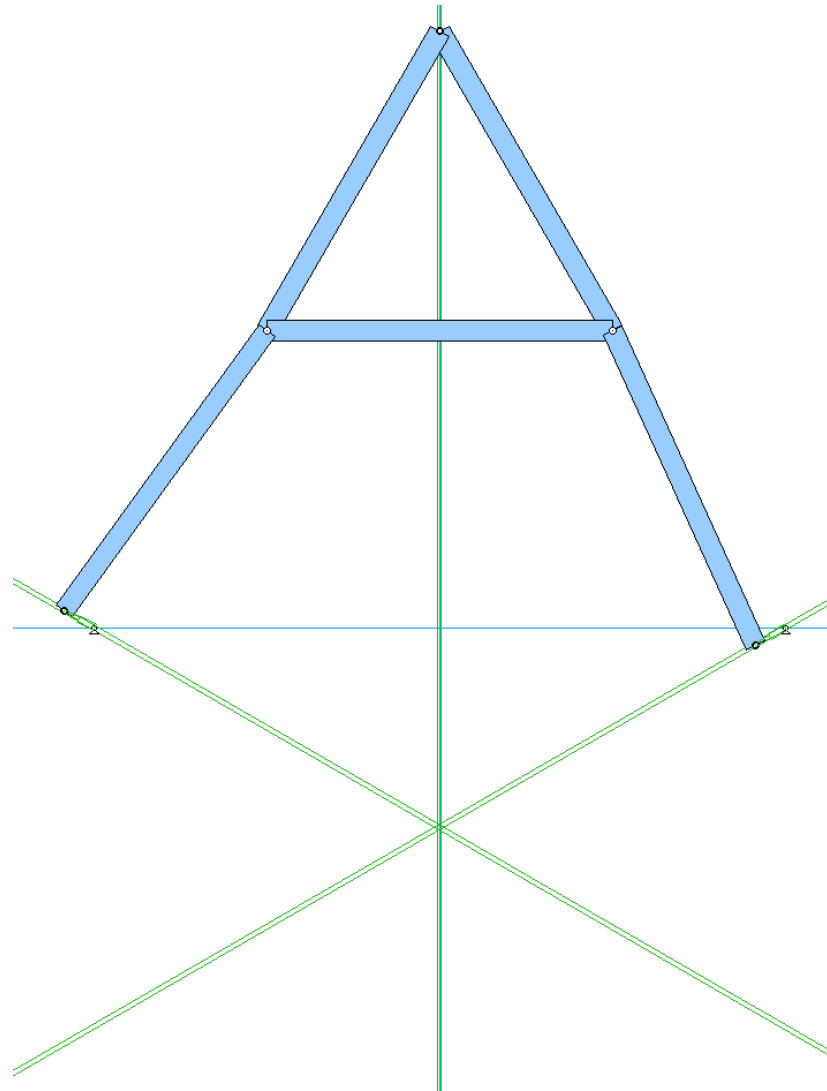
$$\mathbf{J}(\mathbf{q},t)=\begin{bmatrix} c_2 & -c_5 & -r_3s_3 & r_4s_4 \\ s_2 & -s_5 & r_3c_3 & -r_4c_4 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{bmatrix}$$

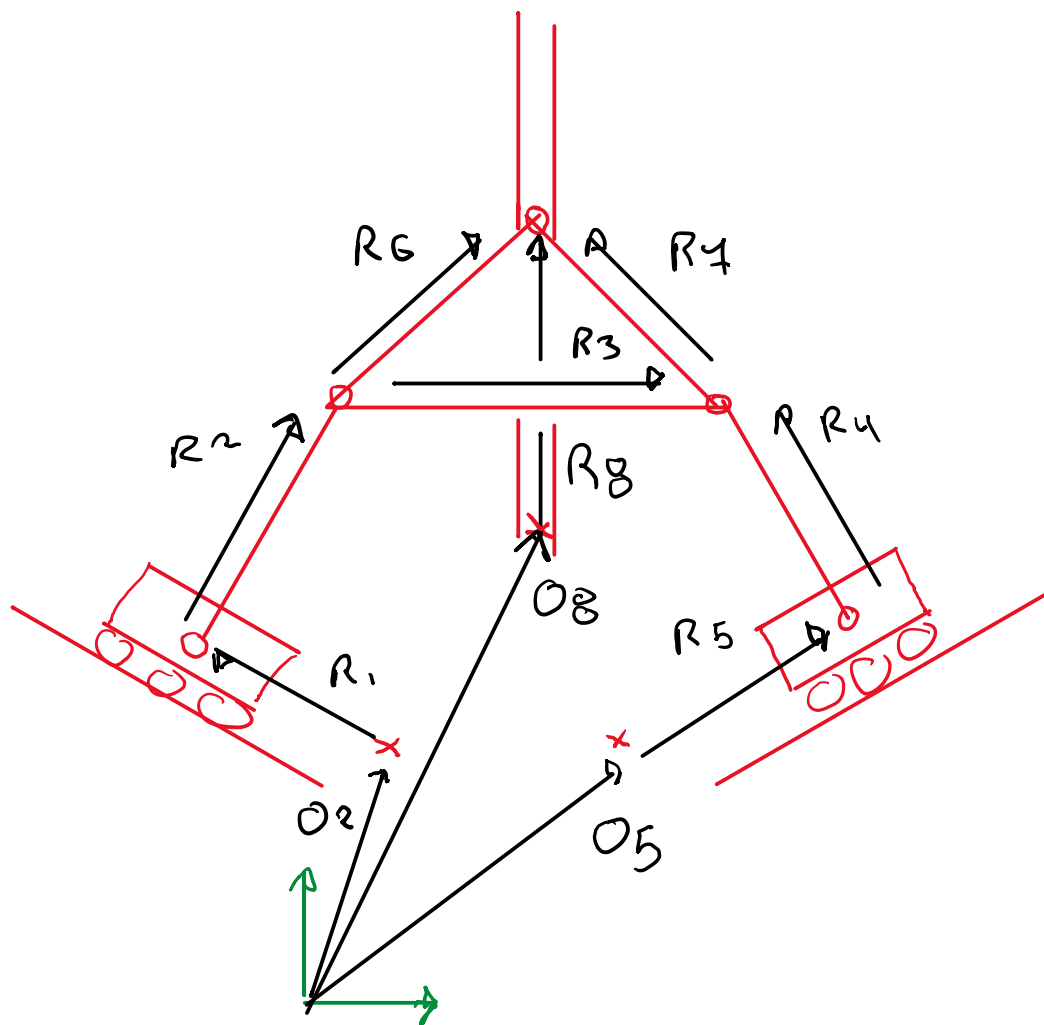
$$\dot{\boldsymbol{\phi}}=\begin{bmatrix} 0 \\ 0 \\ -v_{20}-a_{20}t \\ -v_{50}-a_{50}t \end{bmatrix}$$

$$\dot{\mathbf{J}}(\mathbf{q},t)=\begin{bmatrix} 0 & 0 & -r_3\omega_3c_3 & r_4\omega_4c_4 \\ 0 & 0 & -r_3\omega_3s_3 & r_4\omega_4s_4 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\ddot{\boldsymbol{\phi}}=\begin{bmatrix} 0 \\ 0 \\ -a_{20} \\ -a_{50} \end{bmatrix}$$

# Mecanismo 7 barras triple corredera – 2 GdL





$$\Phi = \begin{bmatrix} \mathbf{o}_2 + \mathbf{r}_1 + \mathbf{r}_2 + \mathbf{r}_3 - \mathbf{r}_4 - \mathbf{r}_5 - \mathbf{o}_5 \\ -\mathbf{r}_3 + \mathbf{r}_6 - \mathbf{r}_7 \\ \mathbf{o}_2 + \mathbf{r}_1 + \mathbf{r}_2 + \mathbf{r}_6 - \mathbf{r}_8 - \mathbf{o}_8 \\ r_1 - r_{10} - v_{10}t - \frac{1}{2}a_{10}t^2 \\ r_5 - r_{50} - v_{50}t - \frac{1}{2}a_{50}t^2 \end{bmatrix} = \begin{bmatrix} o_{2x} + r_1c_1 + r_2c_2 + r_3c_3 - r_4c_4 - r_5c_5 - o_{5x} \\ o_{2y} + r_1s_1 + r_2s_2 + r_3s_3 - r_4s_4 - r_5s_5 - o_{5y} \\ -r_3c_3 + r_6c_6 - r_7c_7 \\ -r_3s_3 + r_6s_6 - r_7s_7 \\ o_{2x} + r_1c_1 + r_2c_2 + r_6c_6 - r_8c_8 - o_{8x} \\ o_{2y} + r_1s_1 + r_2s_2 + r_6s_6 - r_8s_8 - o_{8y} \\ r_1 - r_{10} - v_{10}t - \frac{1}{2}a_{10}t^2 \\ r_5 - r_{50} - v_{50}t - \frac{1}{2}a_{50}t^2 \end{bmatrix} = \mathbf{0} \quad \mathbf{q} = \begin{bmatrix} r_1 \\ \varphi_2 \\ \varphi_3 \\ \varphi_4 \\ r_5 \\ \varphi_6 \\ \varphi_7 \\ r_8 \end{bmatrix}$$

$$\mathbf{J} = \begin{bmatrix} c_1 & -r_2s_2 & -r_3s_3 & r_4s_4 & -c_5 & 0 & 0 & 0 \\ s_1 & r_2c_2 & r_3c_3 & -r_4c_4 & -s_5 & 0 & 0 & 0 \\ 0 & 0 & r_3s_3 & 0 & 0 & -r_6s_6 & r_7s_7 & 0 \\ 0 & 0 & -r_3c_3 & 0 & 0 & r_6c_6 & -r_7c_7 & 0 \\ c_1 & -r_2s_2 & 0 & 0 & 0 & -r_6s_6 & 0 & -c_8 \\ s_1 & r_2c_2 & 0 & 0 & 0 & r_6c_6 & 0 & -s_8 \\ 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \end{bmatrix} \quad \dot{\phi}_t = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ -v_{10} - a_{10}t \\ -v_{50} - a_{50}t \end{bmatrix} \quad \ddot{\phi}_t = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ -a_{10} \\ -a_{50} \end{bmatrix}$$

$$\dot{\mathbf{j}} = \begin{bmatrix} 0 & -r_2\omega_2c_2 & -r_3\omega_3c_3 & r_4\omega_4c_4 & 0 & 0 & 0 & 0 \\ 0 & -r_2\omega_2s_2 & -r_3\omega_3s_3 & r_4\omega_4s_4 & 0 & 0 & 0 & 0 \\ 0 & 0 & r_3\omega_3c_3 & 0 & 0 & -r_6\omega_6c_6 & r_7\omega_7c_7 & 0 \\ 0 & 0 & r_3\omega_3s_3 & 0 & 0 & -r_6\omega_6s_6 & r_7\omega_7s_7 & 0 \\ 0 & -r_2\omega_2c_2 & 0 & 0 & 0 & -r_6\omega_6c_6 & 0 & 0 \\ 0 & -r_2\omega_2s_2 & 0 & 0 & 0 & -r_6\omega_6s_6 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$